COMP 551 Assignment 1: Bike Sharing Regression

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Abstract

In COMP 551 Assignment 1, our team implemented a linear regression machine learning algorithm from scratch to predict the daily number of rented bikes. Using the pandas library, we loaded the dataset and performed systematic data inspection, cleaning, and preprocessing before splitting the data into training (80%]) and testing (20%) subsets. We then implemented multivariable linear regression from the first principles using the closed-form solution (normal equation), and evaluated model performance with mean squared error (MSE) as the loss metric. By comparing training and test MSE, we assessed generalization performance and the reliability of our predictions. In the final stage of the assignment, we explored feature engineering to improve the baseline model's predictive capacity. Specifically, we tested three nonlinear feature transformations applied to the original linear regression framework and compared each approach against the baseline using test-set MSE. This analysis allowed us to determine whether engineered nonlinear features provide improved explanatory power and predictive accuracy for daily bike rental demand.

1 Introduction

Bike-sharing systems have become an increasingly popular component of urban transportation, offering a convenient and low-emission option for short trips and first/last-mile commuting. In Montreal, services such as *BIXI* [1] demonstrate strong adoption and recurring usage patterns. Demand is not constant: rental activity typically varies across seasons and calendar effects (e.g., weekdays versus weekends), and it may differ between casual riders and subscribers. Understanding these trends is valuable because accurate demand forecasting can support operational decisions such as fleet preparation, maintenance planning, and (re)distribution strategies during high-demand periods.

In this assignment, we study bike-rental demand using the *Bike Sharing* dataset [2] dataset from the UCI Machine Learning Repository. The dataset was donated on 12/19/2013 and contains hourly and daily rental counts from 2011–2012 in the Capital Bikeshare system, along with corresponding weather and seasonal information. It also includes separate counts for casual and registered users, which aligns naturally with subscription-based usage analysis. We frame the task as supervised regression: given a set of environmental and calendar-related features, predict the total number of rentals.

As a baseline model, we implemented multivariable linear regression from first principles. We used an 80/20 train–test split and evaluated performance with mean squared error (MSE). We selected the closed-form (normal equation) solution because it is simple to implement, transparent, and provides an interpretable reference point before adding model complexity. Finally, we explored feature engineering by applying three nonlinear transformations to the input features and compared these variants against the baseline using test-set MSE to assess whether nonlinear feature maps improved predictive performance.

2 Data Preprocessing and Exploration

Data Cleaning and Feature Selection The dataset originally consisted of 731 daily records, each containing 16 attributes. To ensure the quality of the linear regression model, we first removed features that were non-informative or redundant.

- **instant & dteday**: Removed as they function solely as unique identifiers or indices and possess no predictive power for general bike demand.
- **temp**: Dropped in favor of **atemp** (apparent temperature). Since these two variables are highly correlated ($R > 0.99$), keeping both would introduce multicollinearity without adding new information. **atemp** was selected as it exhibits a slightly stronger correlation with the target variable and better represents the human perception of weather, which is a key driver in the decision to rent a bike.
- **casual & registered**: Removed to prevent data leakage, as their sum equals the target variable (the total count of users).

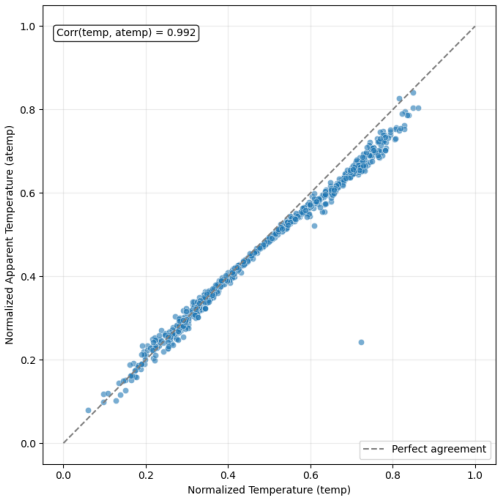


Figure 1: High correlation ($R > 0.99$) between **atemp** and **temp**.

Handling Categorical and Continuous Variables The dataset contained several categorical features: **season**, **weathersit**, **mnth**, and **weekday**. To remove the implied notion of order in these nominal features, we applied *One-Hot Encoding*, which converts categorical variables into multiple binary features. Moreover, for each category, we dropped one binary feature (the reference category) to ensure the validity of the linear regression approach. Since the remaining features can implicitly determine the dropped one, including all of them would introduce perfect multicollinearity, which would render the model's design matrix non-invertible.

To ensure that the model's coefficients were comparable and to improve numerical stability, we applied Standardization to the continuous variables: **atemp**, **hum**, and **windspeed**.

$$x_{new} = \frac{x - \mu}{\sigma} \tag{1}$$

Crucially, statistics (μ, σ) were calculated solely on the training set and then applied to the test set. This protocol prevents information leakage, ensuring the test set remains a strictly unseen proxy.

Data Splitting Strategy The dataset was partitioned into a **Training Set** (80%) and a **Test Set** (20%). A fixed random seed was used to shuffle the indices prior to splitting. This ensures the reproducibility of our results and guarantees that the model is evaluated on a consistent subset of data across different experimental runs.

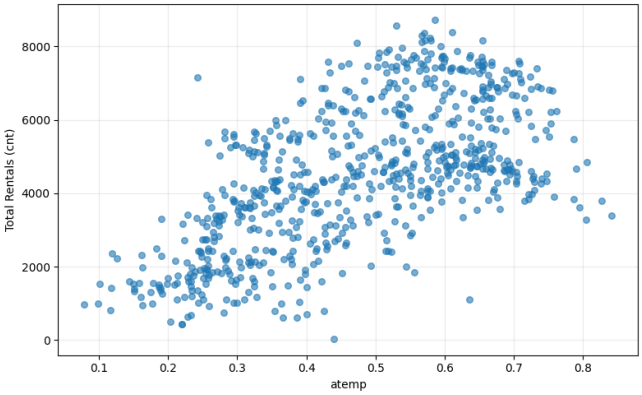
Exploratory Data Analysis Before modeling, we visualized the relationship between the features and the target variable.

The scatter plot in Figure 2a revealed a strong positive correlation, although the relationship appeared quadratic as demand drops in extreme heat. Consequently, we expect the model to assign a significant weight to **atemp**.

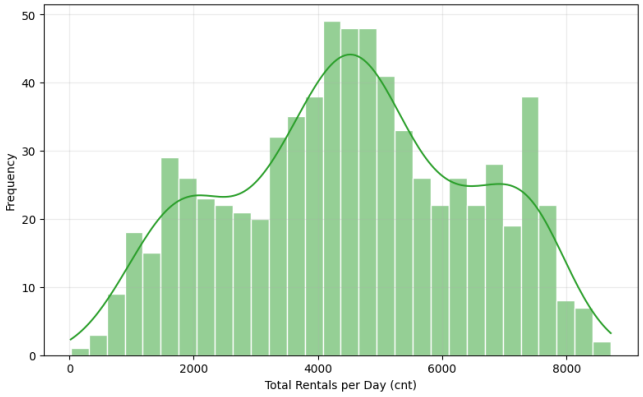
We hypothesized that the error terms would follow a Gaussian distribution. Preliminary visualization of the target variable, as seen in Figure 2b suggested a bell-shaped distribution, supporting the validity of using Least Squares regression.

Figure 2c highlights the importance of retaining the **yr** variable alongside months and seasons, as the market shows a significant general increase in user base during 2012 compared to 2011.

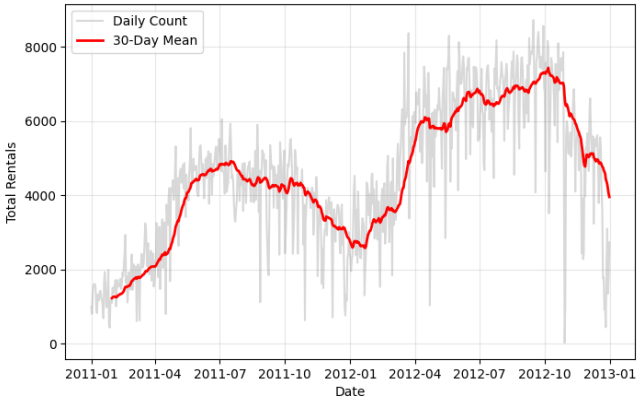
Intuitively, the weather is expected to be one of the most impactful variables in predicting bike rentals. Figure 2d corroborates this, indicating that severe weather conditions have a strong negative impact on user count.



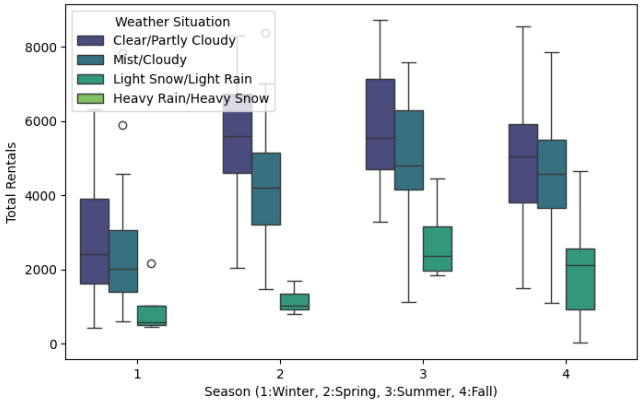
(a) Apparent Temp vs. Rentals



(b) Target Distribution



(c) Yearly Growth (2011-2012)



(d) Weather Severity Impact

Figure 2: Exploratory Data Analysis. (a) Quadratic temperature trend, (b) Normal distribution of target, (c) Yearly market growth, and (d) Negative impact of severe weather.

3 Methods

We model daily bike rental demand as a supervised regression problem and implement multivariable linear regression from scratch using the analytical closed-form solution. After preprocessing, the input features are organized into a design matrix $X \in \mathbb{R}^{n \times d}$, augmented with a column of ones for the intercept, and the targets are collected in a vector $y \in \mathbb{R}^n$, where n is the number of training observations and d is the number of features after one-hot encoding and feature engineering. The model assumes a linear relationship $\hat{y} = Xw$.

Model parameters are estimated by minimizing the mean squared error loss, which yields the normal equation

$$w = (X^\top X)^{-1} X^\top y.$$

For numerical stability, the matrix inverse is not computed explicitly; instead, the equivalent linear system is solved using NumPy's direct solver.

Model performance is evaluated using Mean Squared Error (MSE) on both the training and test sets. All preprocessing steps, including one-hot encoding and standardization of continuous variables, are performed prior to model fitting, with normalization statistics computed solely from the training set to prevent information leakage.

To study the effect of feature engineering on model capacity, we train several variants of the baseline model. These include a polynomial temperature term $\texttt{atemp}^2$, interaction terms of the form $\texttt{atemp} \times \texttt{mnth}_i$, and a logarithmic transformation $\log(\texttt{windspeed} + \texttt{min_value})$. We also evaluate a combined model incorporating both polynomial and interaction features. All variants are trained and evaluated using the same data split and protocol to allow direct comparison of how increased feature complexity affects performance.

4 Results

All models are evaluated using Mean Squared Error (MSE) on both training and test sets to assess fit and generalization.

The baseline multivariable linear regression model achieves a training MSE of 564,867.07 and a test MSE of 626,918.84. As expected, the test error exceeds the training error, reflecting generalization loss and serving as a reference point for feature engineering.

Adding a polynomial temperature term ($\texttt{atemp}^2$) improves performance, reducing the training MSE to 498,012.16 and the test MSE to 585,921.88. Introducing interaction terms between apparent temperature and month ($\texttt{atemp} \times \texttt{mnth}_i$) yields the largest improvement, with a training MSE of 408,419.68 and a test MSE of 515,878.63. A repeated run with an equivalent interaction specification produces nearly identical results, confirming the stability of this improvement. Among all tested models, this interaction-based approach achieves the lowest test-set MSE.

In contrast, applying a logarithmic transformation to windspeed produces negligible change relative to the baseline, with training and test MSE values of 564,549.43 and 626,994.66, respectively. Combining the polynomial and interaction features further reduces training error but slightly worsens test performance compared to the interaction-only model, indicating overfitting due to increased model complexity.

Table 1 summarizes training and test MSE values for all models, enabling direct comparison of feature engineering effects on model performance.

Table 1: Comparison of feature engineering results.

Model Variation	**Train MSE**	**Test MSE**	**Test Improv. (%)**
Baseline (Original)	564,867	626,919	NA
Polynomial ($atemp^2$)	498,012	585,922	-6.54%
Interaction ($atemp \times mnth$)	408,420	515,879	-17.71%
Log-Transform ($wind$)	564,549	626,995	+0.01%
Combined (poly & inter)	408,134	516,101	-17.68%

Residual Analysis Residual diagnostics for the best-performing model are shown in Figure 3. The residual histogram exhibits an approximately symmetric bell shape, and the Q–Q plot shows close alignment with the theoretical normal line, supporting the assumption of normally distributed errors required by Ordinary Least Squares.

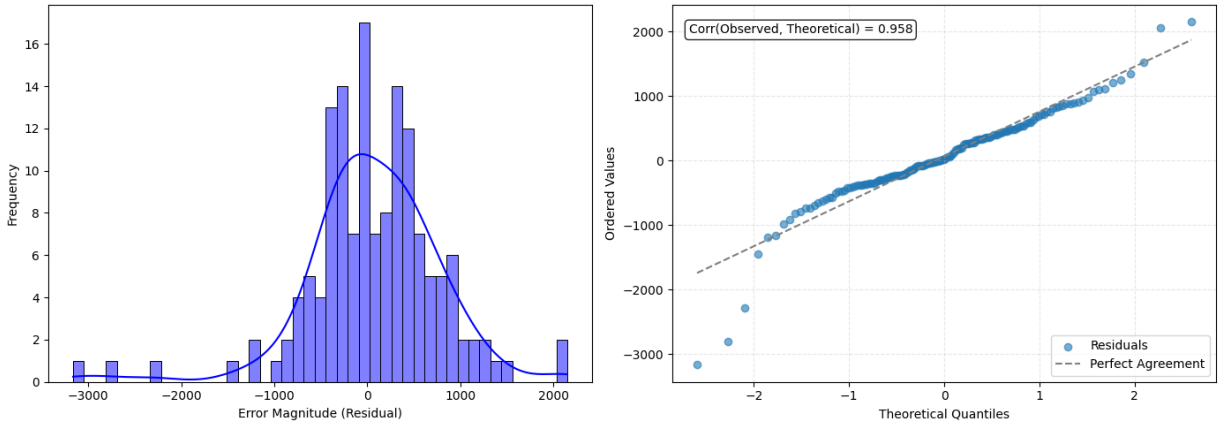


Figure 3: Residual diagnostics for the best-performing model.

5 Discussion and Conclusion

In this project, we applied multivariable linear regression to model daily bike rental demand and examined how preprocessing and feature engineering affect model performance. The baseline model achieved reasonable predictive accuracy, indicating that weather and calendar-related features explain a substantial portion of demand variation. The gap between training and test error suggests that some variability remains unexplained, which is expected given the simplicity of the linear framework and the inherent noise in real-world demand data.

Feature engineering had a clear and measurable impact. Adding a polynomial term for apparent temperature improved performance, supporting the observation that the temperature–demand relationship is not strictly linear. Introducing interaction terms between apparent temperature and month yielded the largest reduction in test error, indicating that temperature effects depend strongly on seasonal context. In contrast, the logarithmic transformation of windspeed produced negligible improvement, consistent with the weak relationship observed during exploratory analysis. Combining polynomial and interaction features further reduced training error but did not improve test performance, indicating overfitting due to increased model complexity.

Overall, this assignment demonstrates that careful preprocessing and targeted feature engineering can meaningfully improve a linear regression model, while also highlighting the trade-off between model expressiveness and generalization. The results illustrate both the usefulness and the limitations of linear methods for modeling real-world data, and underscore the importance of evaluating whether added complexity leads to genuine predictive gains.

6 Statement of Contributions

- **Barry:** Loaded the dataset and supported initial exploration; contributed creative ideas for approaching the assignment tasks; co-wrote and edited the report.
- **Maeve:** Implemented dataset preprocessing and feature inspection; produced visualizations to demonstrate feature correlations and usefulness; led the third part of the assignment (feature engineering with nonlinear modifications); co-wrote and edited the report.
- **Nick:** Implemented the multivariable linear regression model (closed-form/normal equation) and evaluation pipeline; completed the first part of feature engineering; co-wrote and edited the report.

7 Statement of Use of LLM

We used large language model (LLM) tools as *writing and development assistants* in a limited manner. Specifically, we used **ChatGPT** and **Gemini** to improve grammar, clarity, and academic tone, and to suggest LaTeX layout/formatting adjustments. In all cases, we first drafted the report ourselves and then used these tools only to refine wording and presentation; the technical content, experimental design, results, and interpretations remain our own. We reviewed and edited all AI-suggested changes to ensure correctness and consistency with our work.

For programming support, we used LLMs only for **debugging** and for generating **plotting/visualization code** (e.g., graphs for results). Our **linear regression implementation** (including the closed-form solution and evaluation pipeline) was implemented **by hand** and not generated by an AI tool.

References

[1] BIXI Montreal, "Open Data," [Online]. Available: `https://bixi.com/en/open-data/`. [Accessed: Jan. 2026].

[2] H. Fanaee-T and J. Gama, "Bike Sharing Dataset," *UCI Machine Learning Repository*, 2013. [Online]. Available: `https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset`.